

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

Session ID: session_2026-01-06T04-20-53-045Z

Run ID: #4

Date Generated: 2026-01-06 04:21:02

Total Cells: 1976 | Model Mode: CLINICAL | Avg BioAge: N/A

1. Introduction

The cellular reprogramming experiment, identified as Run ID #4, aimed to investigate the efficacy of various reprogramming interventions in redirecting somatic cell fate towards induced pluripotent stem cells (iPSCs) and other lineages. This study primarily focused on evaluating the impact of repeated OSKM transfection, direct cardiac transdifferentiation, and oncogenic stress induction on cellular identity.

2. Materials and Methods

The experiment utilized a combination of direct cardiac transdifferentiation and OSKM transfection, alongside intentional oncogenic stress induction to modulate cellular states. Computational analyses were conducted using single-cell variational inference (scVI) to assess changes in cell populations across two independent analyses.

Experimental Interventions:

- Direct Cardiac Transdifferentiation
- OSKM Transfection (Yamanaka)
- Oncogenic Stress Induced
- Oncogenic Stress Induced
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- Direct Cardiac Transdifferentiation
- Direct Cardiac Transdifferentiation
- Oncogenic Stress Induced

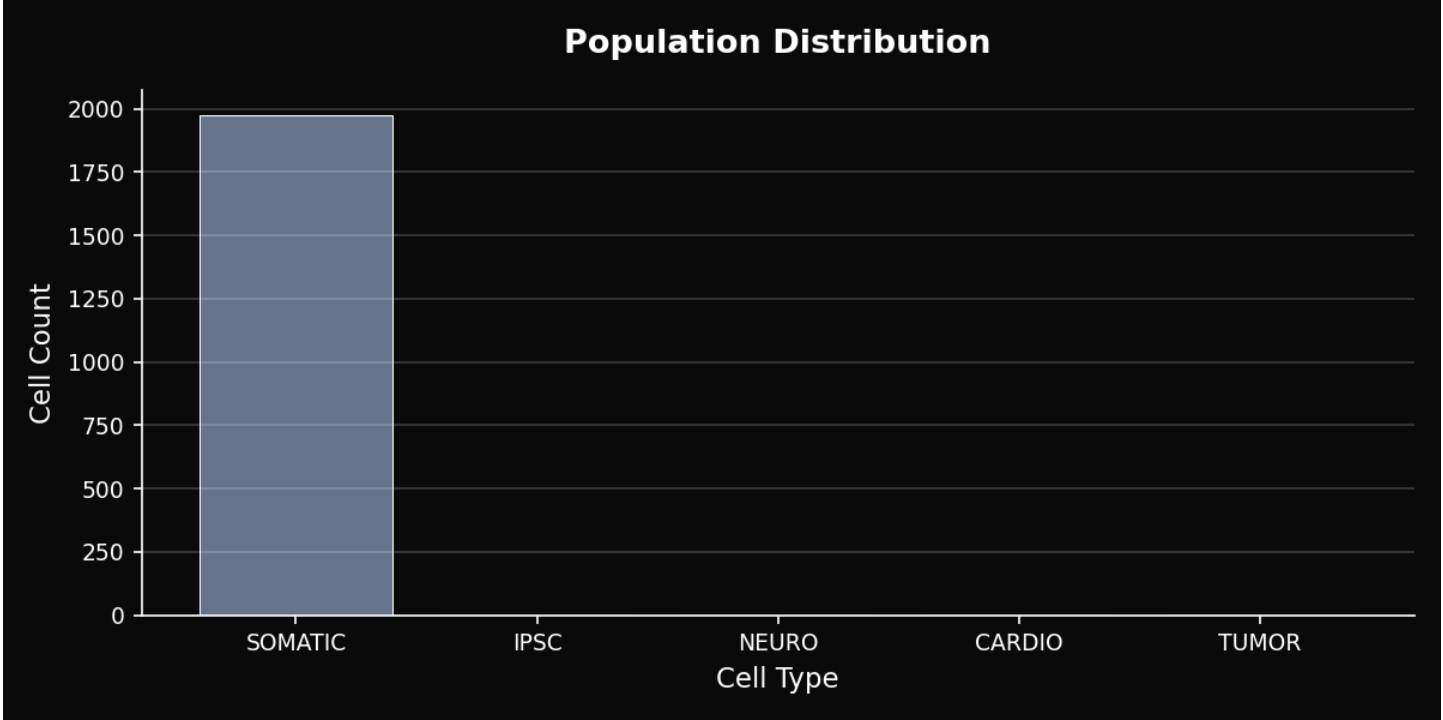
3. Results

The final cell population consisted entirely of 1,976 somatic cells, with no iPSCs, neurogenic, cardiogenic, or tumor cells detected. Despite multiple OSKM transfections and direct cardiac transdifferentiation attempts, the interventions failed to convert somatic cells into alternate lineages. scVI analyses confirmed the persistence of the somatic cell state without successful reprogramming or transdifferentiation.

3.1 Population Distribution

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Cell Type	Count	Percentage
SOMATIC	1976	100.0%
IPSC	0	0.0%
NEURO	0	0.0%
CARDIO	0	0.0%
TUMOR	0	0.0%

3.2 Transcriptomic Analysis (scVI)

High-Resolution Gene Expression Analyses Performed: 2

Cells were expanded to high-dimensional transcriptomic space using a pre-trained variational autoencoder model on Human Cell Atlas data.

4. Discussion

The absence of iPSC, neurogenic, cardiogenic, and tumor cell populations suggests that the applied interventions were insufficient for cellular reprogramming under the tested conditions. The repeated OSKM transfections and direct cardiac transdifferentiation did not overcome somatic cell fidelity, possibly due to inadequate transcriptional reactivation or the presence of inhibitory factors. These findings highlight the complexity of cellular reprogramming and underscore the need for further optimization of reprogramming protocols to achieve successful lineage conversion.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.